INCH-POUND

MIL-M-28787/304A
AMENDMENT 4
28 April 1993
SUPERSEDING
AMENDMENT 3
13 July 1983

MILITARY SPECIFICATION

MODULES, STANDARD ELECTRONIC

MICROPROGRAM CONTROLLER, KEY CODE ZJU

This amendment forms a part of Military Specification MIL-M-28787/304A(EC), dated 10 July 1981, and is approved for use by all Departments and Agencies of the Department of Defense.

PAGES 4, 5, AND 6

*TABLE I: Delete and substitute new table I as printed on pages 3, 4, 5 and 6 of this amendment.

PAGE 7

*4.4.2: Delete and substitute the following:

"4.4.2 Order of testing. Test conditions A through CC shall be performed in any order with the following exceptions. Test conditions A, B, C, D, and E are stress tests and shall be performed only at 25 ± 5°C. If test conditions A, B, C, D, or E are part of a sequence of tests, the order of testing shall be as follows: A, B, C, D, and E where applicable, followed by the remaining tests in the sequence."

*4.5.1, fifth and sixth lines: Delete "C through AA" and substitute "F through AA".

*4.5.3, third line: Delete "C through AA" and substitute "F through AA".

*4.5.3, sixth line: Delete "A through CC" and substitute "F through AA".

PAGE 23

FIGURE 8: Delete and substitute new figure 8 as printed on page 7 of this amendment.

PAGE 31

FIGURE 11: Delete and substitute new figure 11 as printed on page 8 of this amendment.

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PAGES 48 AND 49

TABLE XVIII, Test conditions R13 through R24, Max column: Delete "52" (two places) and substitute "58" (two places).

PAGE 54

TABLE XIX, Test conditions S25 through S27, Max column: Delete "37" (three places) and substitute "44" (three places).

PAGE 56

TABLE XIX, Test condition S52, Max column: Delete "42" and substitute "48".

PAGES 56 AND 57

TABLE XIX, Test conditions S53 through S64, Max column: Delete "38" (two places) and substitute "44" (two places).

NOTE: The margins of this amendment are marked with an asterisk or vertical lines to indicate where changes (additions, modifications, corrections, deletions) from the previous amendment were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous amendment.

Custodians:
Army - ER
Navy - SH
Air Force - 85

Preparing activity:
Navy - NW

(Project 5963-0235)

Review activities:
Army - AT
Navy - AS, MC
Air Force - 13, 19
DLA - ES

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TABLE I. Module electrical requirements.

Test cond	Parameter	Ref test table	Test type 1/-3/	25°C ± 5°C 4/		Class I 4/				Class II 4/				Unit
						0°C ± 5°C		60°C ± 5°C		-55°C ± 5°C		85°C ± 5°C		
				Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	
A	Isolation													
	Initial	III	S	10	INF	10	INF	10	INF	10	INF	10	INF	MOhm
	End-of-life	III		1	INF	1	INF	1	INF	1	INF	1	INF	MOhm
B	Voltage stress check	IV	S	----------Module must pass remaining tests----------										
C1-C22	Input breakdown current	V	S	0.0	1.0	0.0	1.0	0.0	1.0	0.0	1.0	0.0	1.0	mA
D1-D22	Input clamp diode voltage	VI	S	-0.4	-1.5	-0.4	-1.5	-0.4	-1.5	-0.4	-1.5	-0.4	-1.5	V
E1-E16	Output short circuit current	VII	S	-30	-85	-30	-85	-30	-85	-30	-85	-30	-85	mA
F1-F12	Input current, high level	VIII	X	0	80	0	80	0	80	0	80	0	80	µA
F13,F14	Input current, high level	VIII	X	0	30	0	30	0	30	0	30	0	30	µA
F15-F20	Input current, high level	VIII	X	0	40	0	40	0	40	0	40	0	40	µA
F21	Input current, high level	VIII	X	0	50	0	50	0	50	0	50	0	50	µA
F22	Input current, high level	VIII	X	0	100	0	100	0	100	0	100	0	100	µA
G1-G12	Input current, low level	IX	X	0.0	-0.96	0.0	-0.96	0.0	-0.96	0.0	-0.96	0.0	-0.96	mA
G13,G14	Input current, low level	IX	X	0.0	-0.60	0.0	-0.60	0.0	-0.60	0.0	-0.60	0.0	-0.60	mA
G15-G20	Input current, low level	IX	X	0.0	-0.80	0.0	-0.80	0.0	-0.80	0.0	-0.80	0.0	-0.80	mA
G21	Input current, low level	IX	X	0.0	-1.45	0.0	-1.45	0.0	-1.45	0.0	-1.45	0.0	-1.45	mA
G22	Input current, low level	IX	X	0.0	-2.36	0.0	-2.36	0.0	-2.36	0.0	-2.36	0.0	-2.36	mA

- See footnotes at end of table.

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TABLE I. Module electrical requirements - Continued.

Test cond	Parameter	Ref test table	Test type 1/-3/	25°C ± 5°C 4/		Class I 4/				Class II 4/				Unit
						0°C ± 5°C		60°C ± 5°C		-55°C ± 5°C		85°C ± 5°C		
				Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	
H1-H16	Output voltage, high level	X	X	2.4	4.5	2.4	4.5	2.4	4.5	2.4	4.5	2.4	4.5	V
J1-J16	Output voltage, low level	XI	X	0.0	0.5	0.0	0.5	0.0	0.5	0.0	0.5	0.0	0.5	V
K1-K12	High impedance leakage current, high level	XII	X	-1	50	-1	50	-1	50	-1	50	-1	50	µA
L1-L12	High impedance leakage current, low level	XIII	X	1	-50	1	-50	1	-50	1	-50	1	-50	µA
M	Power supply current	XIV	X	28	287	30	303	30	303	34	340	34	326	mA
N1-N10	Dynamic functional test	XV	X	----------		----------		See table XV		----------		----------		
P1-P16	Transition time (tTLH), CL = 50 pF	XVI	S	1	17	1	17	1	19	1	17	1	20	ns
Q1-Q16	Transition time (tTHL), CL = 50 pF	XVII	S	1	10	1	10	1	12	1	10	1	13	ns
R1-R12	Propagation delay time (tPLH), CL = 50 pF	XVIII	S	2	18	2	18	2	21	2	18	2	24	ns
R13-R24	Propagation delay time (tPLH), CL = 50 pF	XVIII	S	5	58	5	58	6	63	5	58	7	71	ns
R25-R27	Propagation delay time (tPLH), CL = 50 pF	XVIII	S	3	37	3	37	4	47	3	37	5	54	ns
R28-R39	Propagation delay time (tPLH), CL = 50 pF	XVIII	S	3	30	3	30	3	39	3	30	4	45	ns

- See footnotes at end of table.

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TABLE I. Module electrical requirements - Continued.

Test cond	Parameter	Ref test table	Test type 1/-3/	25°C ± 5°C 4/		Class I 4/				Class II 4/				Unit
						0°C ± 5°C		60°C ± 5°C		-55°C ± 5°C		85°C ± 5°C		
				Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	
R40-R51	Propagation delay time (tPLH), CL = 50 pF	XVIII	S	3	32	3	32	4	41	3	32	5	47	ns
R52	Propagation delay time (tPLH), CL = 50 pF	XVIII	S	4	42	4	42	5	54	4	42	6	62	ns
R53-R64	Propagation delay time (tPLH), CL = 50 pF	XVIII	S	3	38	3	38	4	49	3	38	5	57	ns
S1-S12	Propagation delay time (tPHL), CL = 50 pF	XIX	S	2	18	2	18	2	21	2	18	2	24	ns
S13-S24	Propagation delay time (tPHL), CL = 50 pF	XIX	S	5	52	5	52	6	63	5	52	7	71	ns
S25-S27	Propagation delay time (tPHL), CL = 50 pF	XIX	S	3	44	3	44	4	47	3	44	5	54	ns
S28-S39	Propagation delay time (tPHL), CL = 50 pF	XIX	S	3	30	3	30	3	39	3	30	4	45	ns
S40-S51	Propagation delay time (tPHL), CL = 50 pF	XIX	S	3	32	3	32	4	41	3	32	5	47	ns
S52	Propagation delay time (tPHL), CL = 50 pF	XIX	S	4	48	4	48	5	54	4	48	6	62	ns
S53-S64	Propagation delay time (tPHL), CL = 50 pF	XIX	S	3	44	3	44	4	49	3	44	5	57	ns
T1-T12	Disable delay time (tPLZ), CL = 50 pF	XX	S	5	55	5	55	6	65	5	55	7	71	ns
U1-U12	Disable delay time (tPHZ), CL = 50 pF	XXI	S	6	60	6	60	7	70	6	60	8	76	ns

- See footnotes at end of table.

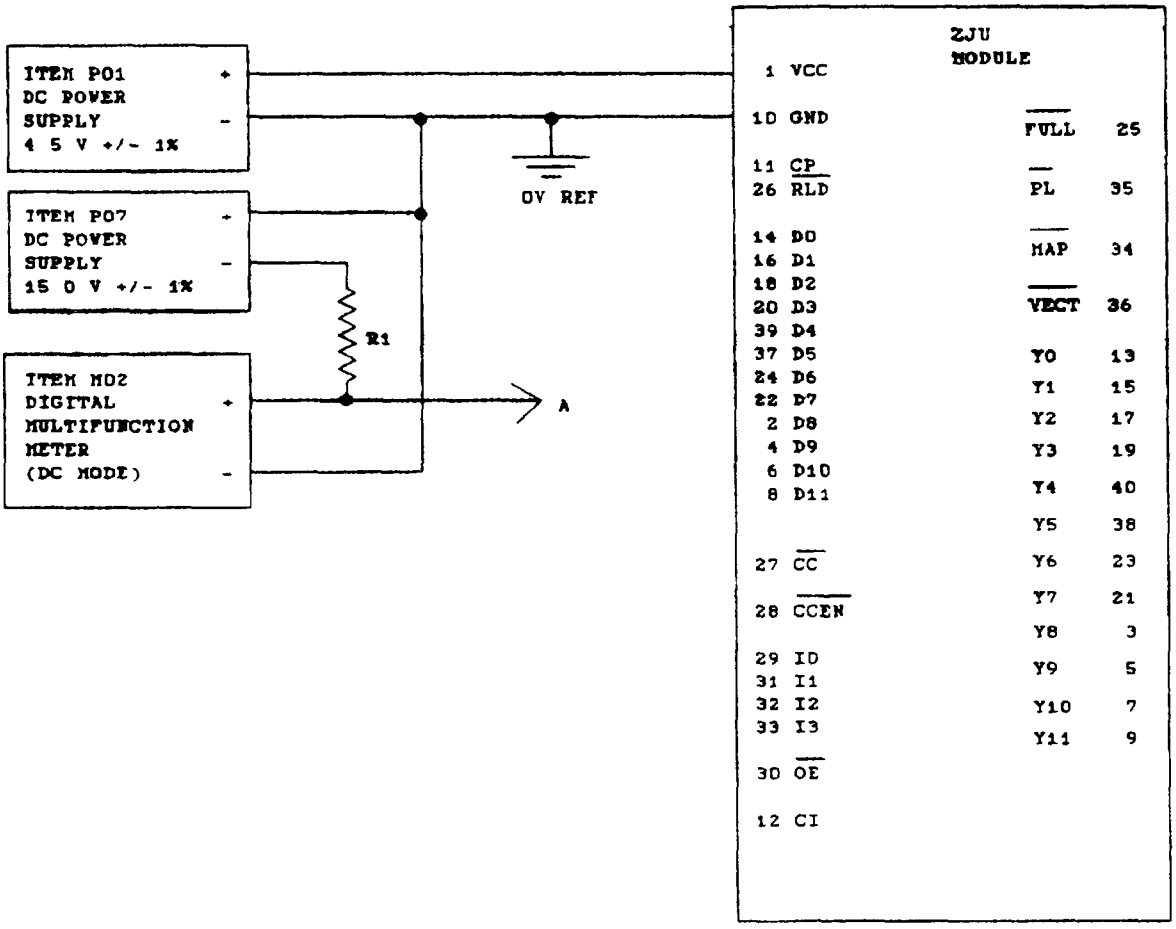
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Test cond	Parameter	Ref test table	Test type 1/-3/	25°C ± 5°C 4/		Class I 4/				Class II 4/				Unit
						0°C ± 5°C		60°C ± 5°C		-55°C ± 5°C		85°C ± 5°C		
				Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	
V1-V12	Enable recovery time (tPZL), CL = 50 pF	XXII	S	5	55	5	55	6	65	5	55	7	71	ns
W1-W12	Enable recovery time (tPZH), CL = 50 pF	XXIII	S	6	60	6	60	7	70	6	60	8	76	ns
X1-X16	Output crosstalk, low level	XXIV	Q	0	1000	0	1000	0	1000	0	1000	0	1000	mV
Y1-Y16	Output crosstalk, high level	XXV	Q	0	500	0	500	0	500	0	500	0	500	mV
Z	Power supply transient amplitude	XXVI	Q	0	400	0	400	0	400	0	400	0	400	mV
AA	Power supply filter capacitance	XXVII	S	3.15	5.85	3.15	5.85	3.15	5.85	3.15	5.85	3.15	5.85	μF
BB1-BB22	Input capacitance	XXVIII	Q	2	30	2	30	2	30	2	30	2	30	pF
CC1-CC16	Output capacitance	XXIX	Q	2	30	2	30	2	30	2	30	2	30	pF

1/ "X" parameters shall be 100 percent tested during module acceptance tests.
2/ "S" parameters shall be sample acceptance tested (1 percent or two modules from each inspection lot, whichever is greater) at 25°C.
3/ "Q" parameters shall be tested during qualification testing only.
4/ Unless otherwise specified, limits are for initial and end-of-life.

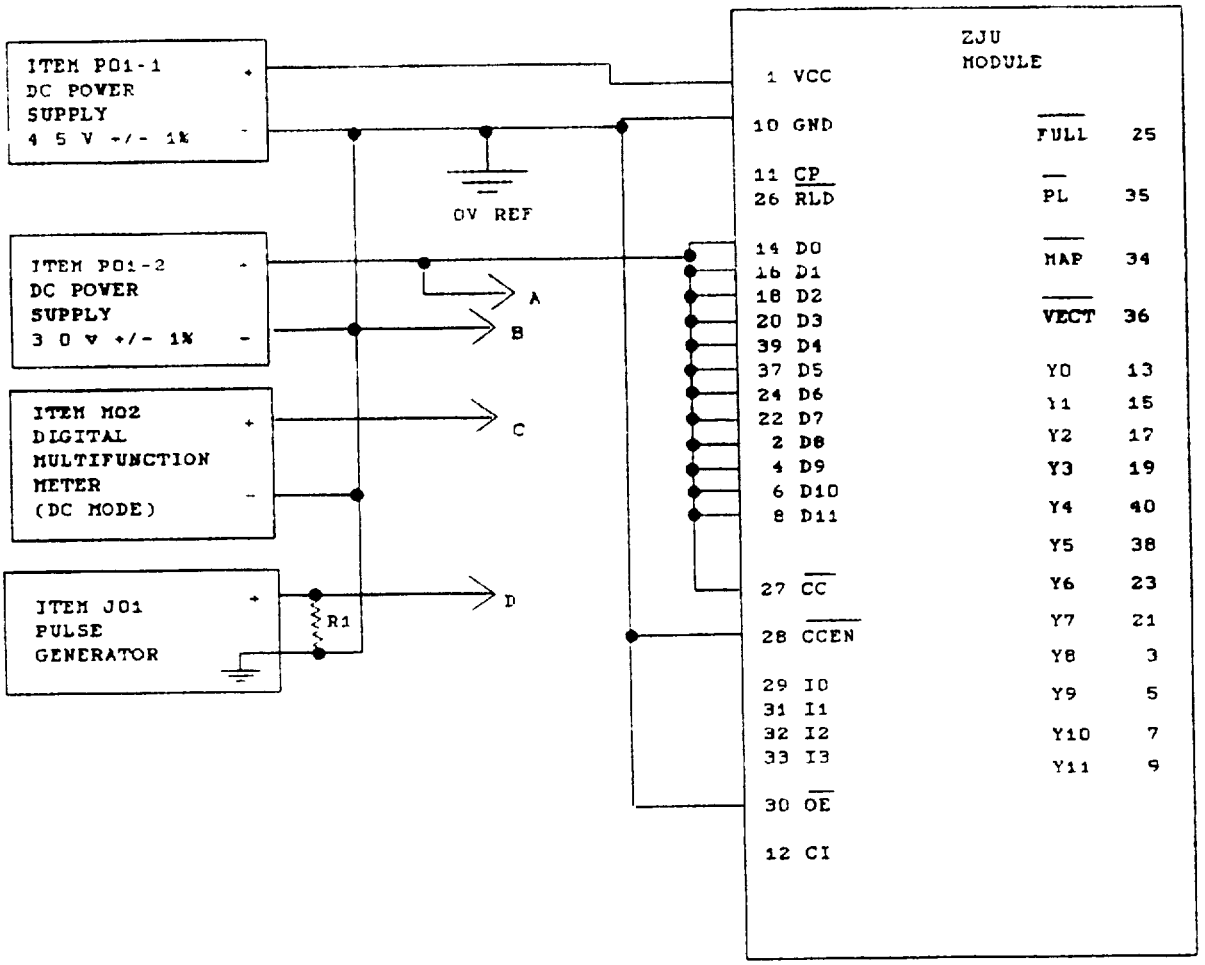
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NOTE: R1 = 750Ω ± 1%, 1/4 W.

FIGURE 8. Input clamp diode voltage test circuit for table VI.

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NOTES:
1. R1 = 51Ω ± 5%, 1/2 W.
2. Load each output per figure 13.
3. Item J01 is used in single pulse mode. In test condition J16, apply 5 pulses as specified on figure 12.

FIGURE 11. Output voltage (high-level and low-level) test circuit for tables X and XI.

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